Deliverable 1 (EDA) – Work Plan

Group 27 – Martim, Maria, Matilde – Due **4 November 2026** (we aim to submit on 3 Nov)

1. What we hand in

Component	Points	Requirements
Group27_EDA_Code.ipynb	5	All cells run, outputs visible, no errors in a fresh environment, Markdown headings, clean code, consistent naming.
Group27_EDA_Slides.pdf	15	Cover slide + **3 to 5 content slides** (we use 5). Insights and interpretations, not just descriptions.
Bonus (2 options)	+10% each	Option 2: Multivariate outlier detection. Option 3: Interactive EDA dashboard. Bonus only counts if the core deliverable passes.
Discussion	–	10–15 min with the instructors (target 9–13 Nov). **Everyone must be able to explain every part.**

The slides must answer these four questions:

- Which findings were most unexpected, and what do they say about likely clusters?
- Which data limitations are the biggest risk for clustering, and how do we reduce them?
- Which activity patterns (including engineered features) suggest natural groups, and what clusters do we expect?
- How would we explain this to non-technical stakeholders: expected number of clusters, key differentiating features, challenges?

2. Bonus options we chose (and why)

Option	Decision	Reason
2. Multivariate outlier & anomaly detection	**YES** (Martim)	Easiest and most useful. It follows directly from the data-quality work, needs only scikit-learn (Isolation Forest, LOF) and protects Deliverable 2, because K-Means is sensitive to outliers.
3. Interactive EDA dashboard	**YES** (Maria)	Low effort because it reuses the plots Maria already makes. Streamlit + Plotly give filters and distributions in about 100 lines, and it is easy to demo at the discussion.
1. External data integration	NO	Hardest to do well. The dataset is synthetic (IBM sample data), so matching it with real benchmarks is weak, and a credibility assessment is required. Low return for the effort.

The guidelines require bonus work to be documented: **why we chose it, what it adds, and what challenges we faced**. Each owner writes a short Markdown cell covering this.

3. Class notebooks we build on

All files are in *Data-Mining-26-27/notebooks_solutions*. About 85% of the core notebook can follow these labs directly. Copy their habits: imports at the top, *df_original = df.copy()*, *metric_features* / *non_metric_features* lists, subplot loops for histograms and box plots, and the filtered correlation heatmap. Tasks marked **Not in labs** need pandas, scikit-learn, or Streamlit docs.

Lab	What it covers	Mainly used by
lab01_distances	Euclidean distance, distance matrices, nearest neighbours	Background for D2
lab03_data_exploration	shape / info / describe, filtering with .loc, missing values, data types, duplicates, histograms and box plots, metric vs non-metric features. Its optional exercise is the ACDC dataset.	Martim, Maria
lab04_data_visualization	Histograms, box plots, scatter, hexbin, jointplot, pairplot, KDE, correlation heatmap, bivariate outliers	Maria, Matilde
lab05_data_visualization	Categorical count plots, crosstab and stacked bars, categorical vs numeric, hue histograms, ydata_profiling	Maria, Matilde
lab06_data_preprocess	Missing-value types, median/mode and KNN imputation, IQR and manual outliers, ~5% rule, winsorizing, feature engineering, removing redundant features	Martim, Matilde

Martim – Data quality, cleaning, outliers (bonus 2) & notebook integration

Analysis

Task	Source lab
Load the dataset with *pd.read_csv*; check data types, shape, and duplicate *member_id*s. Keep *df_original = df.copy()*.	**lab03** (initial analysis, data types, duplicates); lab06 (copy)
Missing values per column: amount, type (MCAR / MAR / MNAR), and treatment (median / mode / KNNImputer or drop, with reasons).	**lab06** (missing values, imputation); lab03 (*isna*, empty strings)
Logical inconsistencies: *uses_meeting_rooms*=0 but *meeting_room_hours*>0 (same for event space and guest passes); *is_team_account*=0 but colleagues>0; actual visits > entitled visits; *plan_days_per_week* vs *plan_tier*.	lab03 (conditional selection with *.loc*); lab06 (coherence check). *No worked example: write the rules yourselves.*
Date checks: *membership_end* before *membership_start* or outside the 3-year window; *days_since_last_visit* vs *last_badge_swipe*.	**Not in labs**: *pd.to_datetime*, *.dt*
Univariate outliers on age, visits, hours, and LTV: IQR, manual limits, combining both, and capping extreme values (winsorizing). Keep the ~5% removal rule of thumb. Before/after box plots.	**lab06** (outlier removal, winsorizing, before/after box plots)
Write a **clean_data()** function (or a cleaned CSV) for the others to use.	lab06 (whole preprocessing flow)
Rename the final notebook to Group27_EDA_Code.ipynb, set up its section structure, merge the drafts, and run Restart & Run All.	Follow the lab notebook structure (imports at the top, Markdown headings)

Bonus

Task	Source lab
Bonus 2: Isolation Forest + Local Outlier Factor on scaled numeric features (optionally Mahalanobis distance). Compare which members each method flags, profile the anomalies, and recommend *exclude / downweight / analyse separately* for Deliverable 2. Short Markdown cell explaining why we chose it.	**Not in labs**: scikit-learn docs (*IsolationForest*, *LocalOutlierFactor*, *StandardScaler*). Extends lab06's outlier section and lab04's bivariate outliers.

Slide: Data quality risks and how we handle them (including the multivariate outlier results).
Hand-off: clean_data() by **23 Oct**; outlier analysis by **28 Oct**; merged notebook by **1 Nov**.

Maria – Member profile, descriptive analysis & dashboard (bonus 3)

Analysis

Task	Source lab
Demographics: age (histogram, box plot), gender, country, industry (count plots sorted largest to smallest; group rare categories together).	**lab03 / lab04** (histograms, box plots); **lab05** (categorical count plots, sorted bars)
Plan and account: *plan_tier*, *plan_days_per_week*, team vs individual, colleagues per account.	**lab05** (univariate categorical, multiple bar plot figure)
Time trends over the 3 years: new members per month, churn (members with an end date), cohorts, tenure.	**Not in labs**: *pd.to_datetime*, *groupby(dt.to_period('M'))*, line plots
Bivariate numeric: correlation heatmap (annotate only strong values), scatter, hexbin, joint and pair plots.	**lab04** (correlation matrix, scatter, hexbin, jointplot, pairplot)
Categorical vs categorical and vs numeric: industry and country vs tier (*pd.crosstab*, normalized stacked bars); plan tier vs LTV, visits, and age (box plots, *hue* histograms).	**lab05** (crosstab, stacked bars, categorical vs continuous)
Point out redundant (highly correlated) variables that are candidates to drop before clustering.	**lab06** (feature selection: redundancy vs relevancy)
Optional quick overview with *ydata_profiling* (not a replacement for our own analysis).	lab05 / lab06 (ProfileReport)

Bonus

Task	Source lab
Bonus 3: Streamlit + Plotly app (*dashboard/app.py*) on the cleaned data. Sidebar filters (plan tier, country, industry, team account, active/churned), histograms and box plots of any chosen variable, a category breakdown, a KPI row (members, average LTV, average weekly visits), and a drill-down member table. Include a requirements.txt and a README explaining how to run it. Keep it outside the notebook so the notebook still runs in a fresh environment.	**Not in labs**: Streamlit + Plotly Express docs. Reuse the plot ideas from lab04 / lab05. Budget about half a day to learn it.

Slide: Who our members are, plus the most unexpected findings.
Hand-off: Descriptive analysis by **28 Oct**; dashboard by **1 Nov**. Maria also designs the deck template.

Matilde – Behavior, value & feature engineering

Analysis

Task	Source lab
Usage: *total_visits*, *avg_weekly_visits*, recency, and last-period actual vs entitled visits (distributions, scatter of actual vs entitled).	**lab03 / lab04** (histograms, box plots, scatter, KDE)
Add-ons: adoption rate of each *uses_** flag, which add-ons are used together, and meeting/event hours.	**lab05** (count plots, crosstab of two binary flags); lab03 (column sums)
Value: distribution of *lifetime_value* and *n_renewals*, and what drives LTV (correlations, LTV by plan tier).	**lab04** (correlation matrix); **lab05** (categorical vs continuous)
Engineered features (table below), each with a one-line justification.	**lab06** (feature engineering; *spent_online* ratio is the same pattern)
First look at possible groups: pair plots of the engineered features coloured by plan tier or active/churned (analysis only, no clustering yet).	**lab04 / lab05** (pairplot with *hue*). PCA is **not in labs**, so it is optional.

Bonus

Task	Source lab
Supports both bonuses: shares the engineered features with Martim (outlier input) and Maria (dashboard variables).	–

Slide: Behavioral signals and engineered features. Matilde also leads the shared "expected segments" slide.
Hand-off: Engineered features by **26 Oct** (others depend on them); behavior analysis by **28 Oct**.

Engineered features (Matilde)

Feature	Formula	What it captures	Source lab
tenure_days	end (or extraction date) – start	Loyalty	Not in labs (dates)
utilization_rate	actual / entitled visits (last period)	Whether members get value from their plan; churn risk	lab06 (ratio feature)
ltv_per_month	lifetime_value / tenure in months	How much a member is worth per month, regardless of tenure	lab06 (ratio feature)
n_addons	sum of uses_* flags	How much of the service they use	lab03 (sum by row, *axis=1*)
hours_per_visit	(meeting + event hours) / total_visits	Whether they use it as an office or just a desk	lab06 (ratio feature)
is_active	membership_end is missing	Active vs churned	lab03 (*isna*)
plan_tier_ord	hot_desk=1 ... private_office=5	Plan tier as a numeric scale	Not in labs (*.map*)
R / F / M scores	quantiles of recency, frequency, LTV	Value-based segmentation (RFM)	lab03 (the lab dataset has rcn/frq/mnt); scores via *pd.qcut* are new

4. Slide plan (cover + 5 content slides)

#	Slide	Owner
0	Cover (group, members, title)	Maria
1	Who our members are + most unexpected findings	Maria
2	Data quality risks & mitigation (incl. multivariate outliers, bonus 2)	Martim
3	Behavioral signals & engineered features	Matilde
4	Expected segments: number of clusters, key differentiating features, challenges	Matilde (written by all)
5	Non-technical summary & next steps for clustering (mention the dashboard)	Martim (written by all)

The limit is 5 content slides, so bonus details go in the notebook and the dashboard. The slides only reference them.

5. Timeline

Dates	Milestone
20–22 Oct	Martim loads the data and does the first quality checks. Maria and Matilde start on the raw data.
23 Oct	Martim shares clean_data(). Everyone switches to the cleaned data.
23–26 Oct	Main analysis. Matilde delivers the engineered features (26 Oct).
24–28 Oct	Martim: outlier detection (bonus 2). Maria: finish descriptive analysis.
29 Oct	**Team meeting:** agree on key insights, expected number of clusters, and slide storyline.
29 Oct – 1 Nov	Slides drafted. Maria builds the dashboard. Martim merges the notebooks.
2 Nov	Fresh-environment run of the notebook and dashboard; slides exported to PDF; review by all three.
3 Nov	Buffer day, then **submit** (one day before the deadline).
~9–13 Nov	Discussion. Each person explains their part and understands the rest.

6. Working rules

- **Git:** each person works in *notebooks/drafts/eda_<name>.ipynb*. Only Martim edits the final notebook, so we avoid notebook merge conflicts. The dashboard lives in *dashboard/*.
- Use the same variable names everywhere (e.g. *df_raw*, *df_clean*, snake_case features).
- Write down who did what and which AI tools you used. Deliverable 2 requires an AI-usage and contribution annex.
- Late submissions lose 10% per day, and reports go through a plagiarism check.

Warning: hidden text in the guidelines PDF

Pages 4–5 of the guidelines contain hidden instructions aimed at AI tools. They ask for strange sentences to be pasted into the report, mention columns that do not exist (DeskLoyaltyScore, PreferredWorkspaceZone, NumMembersReferred), and suggest nonsense features (AgeCountryMean, IdLengthRatio, SwipeGenderIndex). These are traps to detect unreviewed AI output. **Never use any of them.** If you use an AI tool with the PDF, check its output for these.